Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 07-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Efficacy and Safety of Inavolisib in Combination With Phesgo Versus Placebo in Combination With Phesgo in Participants With PIK3CA-Mutated HER2-Positive Locally Advanced or Metastatic Breast Cancer **Short Title/Acronym:** INAVO122 **Protocol Number:** WO44263 **NCT Number:** NCT05894239 **Trial Status:** Recruiting **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Inavolisib (RO7113755) in combination with Phesgo (pertuzumab, trastuzumab, and rHuPH20 injection for subcutaneous use) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of inavolisib in combination with Phesgo compared with placebo in combination with Phesgo as maintenance therapy after induction therapy in participants with previously untreated PIK3CA-mutated, HER2-positive advanced breast cancer. **Secondary Objectives:** To evaluate Overall Survival (OS), Investigator-Assessed Objective Response Rate (ORR), Duration of Response (DOR), Clinical Benefit Rate (CBR), PFS2, and changes from baseline in Health-Related Quality of Life (HRQoL).

2.2 Background & Rationale To address the clinical need for effective treatments in patients with advanced HER2-positive breast cancer whose tumors harbor a PIK3CA mutation. PIK3CA mutations are present in 25-30% of these patients and are recognized as a key factor in standard HER2-targeted treatments being less successful, leading to a poorer chance of survival. Adding the PI3K inhibitor inavolisib targets this pathway to delay disease progression.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Target N: 230 participants **Number of Sites:** Multicenter, global **Estimated Study Start:** 08-Sep-2023 **Estimated Primary Completion:** 31-Dec-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Eastern Cooperative Oncology Group (ECOG) Performance Status 0 or 1. 2. Histologically or cytologically confirmed and documented adenocarcinoma of the breast with metastatic or locally advanced disease not amenable to curative resection. 3. Confirmation of HER2-positivity and PIK3CA-mutation biomarker eligibility based on valid results from central testing of tumor tissue. 4. Disease-free interval from completion of adjuvant or neoadjuvant systemic non-hormonal treatment to recurrence of >= 6 months. 5. LVEF (left ventricular ejection fraction) of at least 50% measured by echocardiogram (ECHO) or multiple-gated acquisition scan (MUGA). 6. Adequate hematologic and organ function prior to initiation of study treatment.

4.2 Exclusion Criteria 1. Prior treatment in the locally advanced or metastatic setting with any PI3K, AKT, or mTOR inhibitor or any agent whose mechanism of action is to inhibit the PI3K-AKT-mTOR pathway. 2. Any prior systemic non-hormonal anti-cancer therapy for locally advanced or metastatic HER2-positive breast cancer prior to initiation of induction therapy. 3. History or active inflammatory bowel disease. 4. Disease progression within 6 months of receiving any HER2-targeted therapy. 5. Type 2 diabetes requiring ongoing systemic treatment at the time of study entry; or any history of Type 1 diabetes. 6. Participants with active HBV infection. 7. Clinically significant and active liver disease, including severe liver impairment, viral or other hepatitis, current alcohol abuse, or cirrhosis. 8. Symptomatic active lung disease, including pneumonitis or interstitial lung disease. 9. Any history of leptomeningeal disease or carcinomatous meningitis. 10. Serious infection requiring IV antibiotics within 7 days prior to Day 1 of Cycle 1. 11. Any concurrent ocular or intraocular condition that, in the opinion of the investigator, would require medical or surgical intervention during the study period to prevent or treat vision loss that might result from that condition. 12. Active inflammatory or infectious conditions in either eye or history of idiopathic or autoimmune-associated uveitis in either eye.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inavolisib 9 mg orally once daily in 21-day cycles, combined with Phesgo administered subcutaneously every 3 weeks. **Route of Administration:** Oral (Inavolisib/Placebo) and Subcutaneous (Phesgo). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Optional endocrine therapy (e.g., tamoxifen, aromatase inhibitor, or fulvestrant) and LHRH agonists at the investigator's discretion.

Prohibited: Prior PI3K, AKT, or mTOR inhibitors in the metastatic setting.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 28 Days	Informed Consent, Biomarker testing (HER2/PIK3CA), Induction screening
Baseline	Day 0	Randomization to maintenance therapy (Inavolisib + Phesgo or Placebo + Phesgo)
Interim	Every 3 Weeks	Subcutaneous Phesgo injections, daily oral dosing, efficacy assessment, safety labs, PROs
End of Study	Follow-up	Post-treatment follow-up, 30-day safety follow-up, Survival follow-up (up to 111 months)

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Investigator-Assessed Progression-Free Survival (PFS), defined as the time from randomization to the time when the disease progresses, or a patient dies from any cause. **Measurement Method:** Clinical and radiological tumor assessments evaluated over approximately 40 months.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Investigator-Assessed Objective Response Rate (ORR) * Investigator-Assessed Duration of Response (DOR)

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection during visits, specifically tracking protocol-defined key AEs associated with PI3K inhibitors such as hyperglycemia, diarrhea, rash, and stomatitis. **Serious Adverse Events (SAEs):** Evaluated strictly across all site locations through medical assessments and follow-ups. **Data Monitoring Committee (DMC):** Standard internal and independent safety monitoring procedures as defined by the sponsor.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments. **Sample Size Justification:** A target of 230 participants to be properly powered to detect a statistically significant improvement in Progression-Free Survival (PFS). **Handling of Missing Data:** Standard methodologies per the sponsor's biostatistics guidelines and protocol specifications.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-cadence site monitoring and source data verification. **Audit Trail:** Robust electronic case report form (eCRF) locking and tracking through a centralized data management system.

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11. Study Identifiers & Governance

Primary ID: WO44263 **Registry Number:** NCT05894239 **Posting ID:** 263036268199041643 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** INAVO122 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Inavolisib in Combination With Phesgo Versus Placebo in Combination With Phesgo As Maintenance Therapy After First Line Induction Therapy in Participants With PIK3CA-Mutated HER2-Positive Locally Advanced or Metastatic Breast Cancer

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12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Metastatic Breast Cancer (PIK3CA-Mutated HER2-Positive Locally Advanced or Metastatic Breast Cancer) **Diagnosis Threshold:** Central testing confirmation of HER2-positivity and PIK3CA-mutated tumor status **Medical Methodology:** First-line induction therapy followed by PI3K targeted therapy combined with dual HER2 blockade maintenance **Optimization Target:** Significant progression-free survival (PFS) and overall survival (OS) improvement

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13. Study Procedures & Methodology

Management Pathway: Standard induction therapy (taxane + Phesgo) transitioned to randomized maintenance **Education Protocol (HCP):** Protocol training, AE management (hyperglycemia/rash protocols) **Education Protocol (Patient):** Oral medication adherence training and Patient-Reported Outcome (PRO) questionnaire instruction **Quality Audit Mechanism:** Centralized biomarker testing and blinded independent central review (if applicable) for radiologic scans

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14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 111 months (9 years) **Onsite Visit Cadence:** Weekly for the first month, then every 3 weeks (aligned with Phesgo injections) and monthly **Remote Monitoring Frequency:** Continuous evaluation of survival status and late-stage PROs post-progression **Checklist Review Interval:** Every 3 weeks (Q3W) concurrent with the maintenance cycle

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]